Impact of Utility-Scale Solar Development on Land Use and Carbon Dynamics in Massachusetts

Selected Dissertation Research Slides

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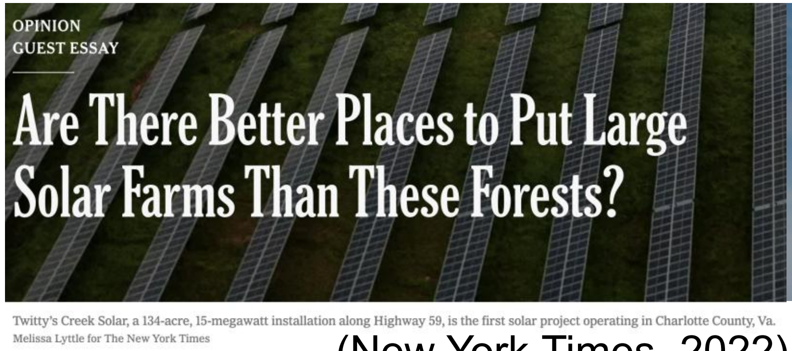
Earth and Environment

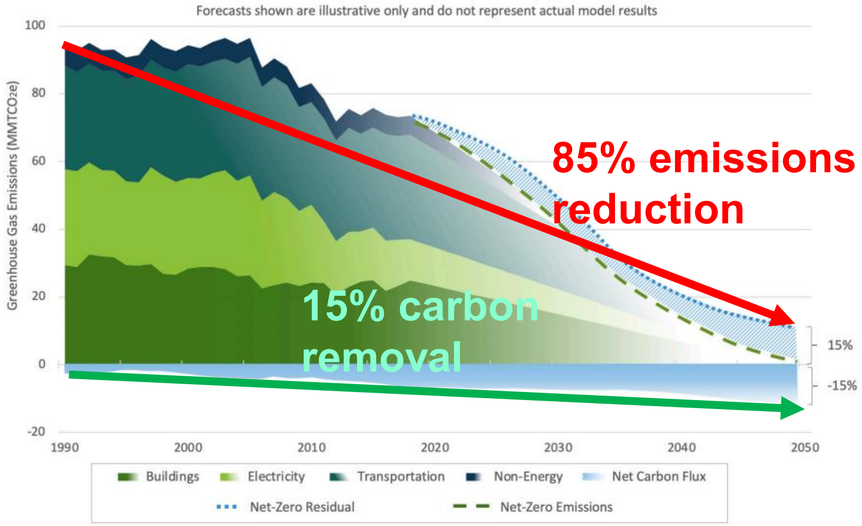
Solar expansion is contributing to forest loss

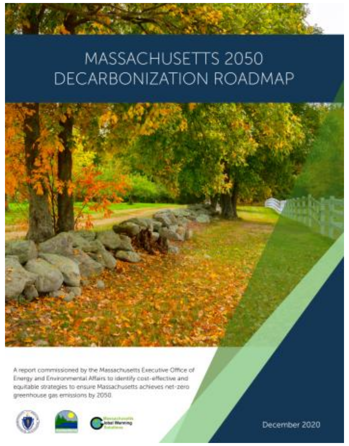
- Utility-scale solar requires land (Ong et al., 2013)
- Solar development cause deforestation and ecosystem changes (Hernandez et al., 2014)
- 1,519 ha of forest cleared for ground solar in Massachusetts (Manion et al., 2023)
- Massachusetts aims carbon neutral through 85% reductions and 15% removals by 2050 (MASS EEA, 2020)



(New York Times, 2022)

(Manion et al., 2023)





(MASS EEA, 2020)

Achieving net zero requires both clean energy expansion and protection of terrestrial carbon sinks

Research objectives and questions

Overall objective

Quantify the net carbon impacts of utility-scale solar development in Massachusetts

How the net carbon impacts vary through time?

$$C_{net}(t) = C_{Deforestation}(t) + C_{LCA}(t) - C_{Saving}(t)$$

Objective 1 – Quantify solar-associated land cover change (Chapter 1)

How can long-term solar-associated deforestation be monitored and quantified?

Objective 2 – Energy system carbon impacts (Chapter 2)

How much carbon is emitted and avoided from solar electricity at different times?

Objective 3 – Integrated net carbon modeling with biomass (Chapter 3)

To what extent do deforestation emissions offset solar carbon benefits?

Chapter 1.

Estimating the Area of Utility-Scale Solar and Associated Deforestation between 2005-2024 and Their Uncertainties

Mapping solar development from Landsat time series

CCDC-SNIC

Temporal Segmentation (CCDC) → Spatial Segmentation (SNIC) → Change Classification → Area Estimation

Temporal Segmentation (CCDC)

- Landsat time series 20 years (2005-2024)
- Entire extent of Massachusetts
- Input bands
  NDVI, NDBI (Building), BSI (Bare Soil), NDTI (Tillage), Albedo, LST (Temperature)

Spatial Segmentation (SNIC)

- Features are selected from CCDC output
- SNIC generated spatial segmentation relevant to utility-scale solar development

Change Classification

- Random Forest classifier
- Change Stratum
  1) Deforestation near solar panels
  2) Solar panels after deforestation
  3) Other solar panels (No deforestation)
  4) Other land changes

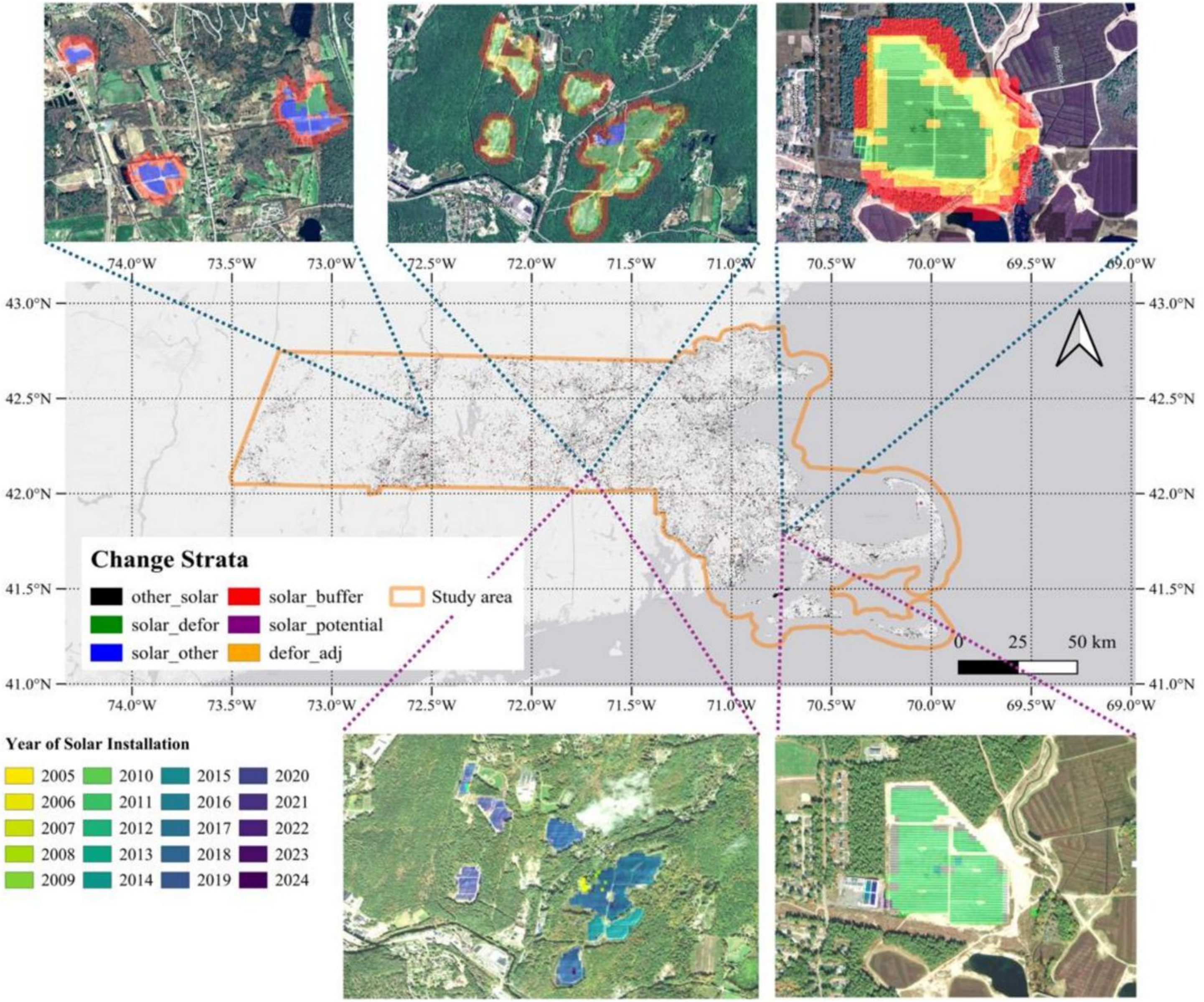
Area Estimation

- Stratified random sampling
- Find possible omission errors
- Area and uncertainty estimation
- Correct bias / provide uncertainty bounds

Mapped area ≠ True area !

Provide statistically viable area estimation for carbon modeling

Land cover change results



Area estimates of target classes

Merging change classes

Reference Map	Solar after deforestation	Deforestation near solar	Other solar panels	Other land change
Area Estimates (km^2)	21.104	13.521	20.391	1372.253
Map Area (km^2)	19.85	14.52	17.71	1375.19
Area Standard Errors (km^2)	3.006	2.533	4.104	5.448
95% Confidence Interval (km^2)	5.892 (27.9%)	4.965 (36.7%)	8.045 (39.5 %)	10.678

Reference Map	Solar-driven deforestation	Solar panels
Area Estimates (km^2)	34.626	41.495
Map area (km^2)	34.370	37.560
Area Standard Errors (km^2)	3.751	5.026
95% Confidence Interval (km^2)	7.352 (21.2%)	9.852 (23.7%)

- Solar-associated deforestation increased **by 66%** when accounting for buffer deforestation (21 → 35 km^2)
- Area estimates provide key inputs for carbon modeling (Solar-driven deforestation: 35 ± 7 km^2, Solar panels 41 ± 10 km^2)
- Area uncertainties remained relatively low despite small target classes (Relative uncertainty: 21.2 / 23.7 %)

Key takeaways

- Developed a CCDC-SNIC framework to map solar expansion and associated land cover change
- Utility-scale solar development covers less than **0.15%** of Massachusetts
- About **half (51%)** of utility-scale solar development in Massachusetts occurred on deforested land
- Forest-based projects require **66% more clearing** than the solar footprint (**1.66 ha forest loss per 1 ha solar**)
- These area estimates provide the key inputs for net carbon impact assessment of utility-scale solar

Cho, K. & Woodcock, C. E. Detecting Utility-Scale Solar Installations and Associated Land Cover Changes Using Spatiotemporal Segmentation of Landsat Imagery. *Science of Remote Sensing*. 2025, p.100337.

Chapter 2.

Carbon Impacts of Utility-Scale Solar
: Life Cycle Emissions and Carbon Savings

$$C_{net}(t) = C_{Deforestation}(t) + \boxed{C_{LCA}(t) - C_{Saving}(t)}$$

Overview of solar energy system modeling

Life cycle emissions from utility-scale solar

- Life cycle emissions (Carbon intensity) from PV life cycle assessment literature
- System assumptions from NREL report

An Updated Life Cycle Assessment of Utility-Scale Solar Photovoltaic Systems Installed in the United States

Brittany L. Smith, Ashok Sekar, Heather Mirletz, Garvin Heath, and Robert Margolis

Carbon saving from displacement Scenario

- Solar has lower carbon intensity than conventional electricity
- Replacing higher-carbon electricity creates avoided emissions
- Carbon savings depend on the displacement source
- Displacement scenarios
  Natural gas (Upper)
  ISO-NE grid (Lower)

Solar electricity generation

- Solar generation estimated using NREL SAM
- Converts carbon intensity into total system carbon impacts

System Advisor Model

Energy system carbon modeling

Life Cycle Emissions from Solar

$$C_{LCA} = \boxed{EF_{solar}} \times A_{solar} \times \boxed{\frac{1}{\alpha} \times \frac{E_{SAM}}{100}} \times \frac{12}{44} \times 10^{-6}$$

$$[Mg\ C] = \left[\frac{g\ CO_2 eq}{kWh} \times ha \times \frac{MW_{dc}}{ha} \times \frac{kWh}{MW_{dc}} \times \frac{C}{CO_2 eq} \times \frac{Mg}{g}\right]$$

Carbon savings replacing carbon Intensity of the grid

$$C_{Saving} = \boxed{EF_{grid}} \times A_{solar} \times \boxed{\frac{1}{\alpha} \times \frac{E_{SAM}}{100}} \times \frac{12}{44} \times 10^{-6}$$

$$[Mg\ C] = \left[\frac{g\ CO_2 eq}{kWh} \times ha \times \frac{MW_{dc}}{ha} \times \frac{kWh}{MW_{dc}} \times \frac{C}{CO_2 eq} \times \frac{Mg}{g}\right]$$

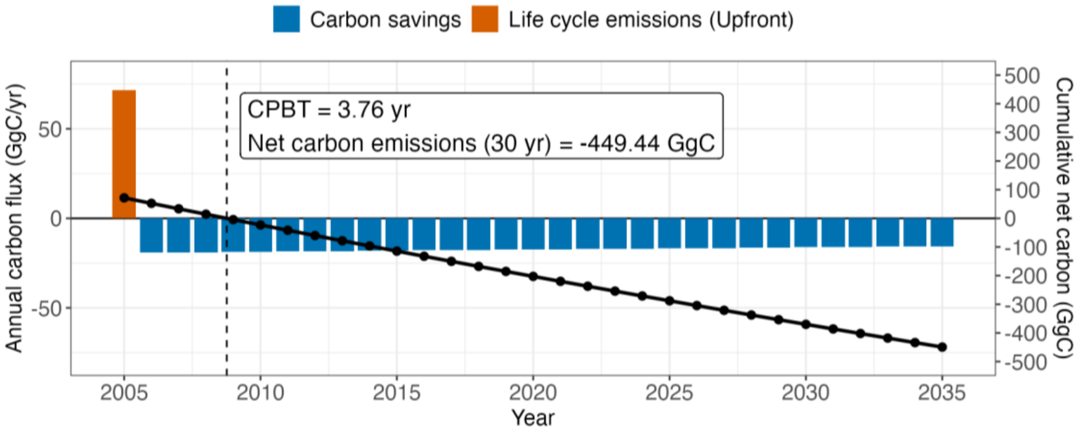
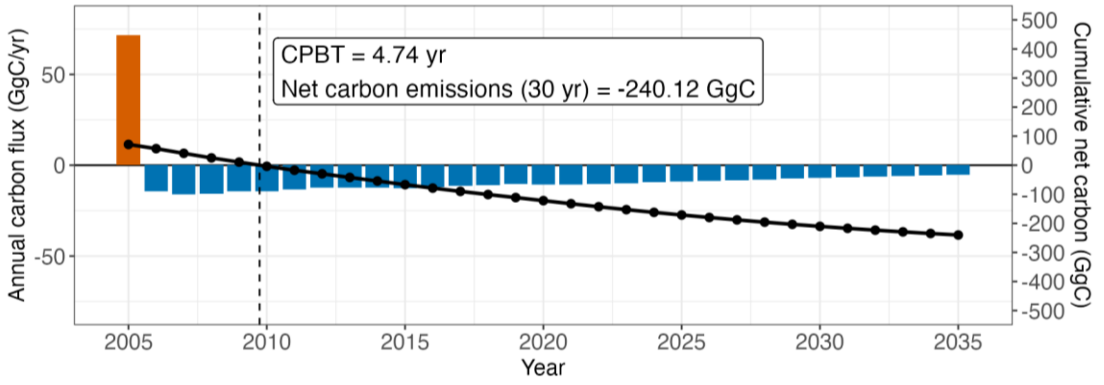
Where:
EF_{solar} : Emission factor from LCA analysis of utility-scale solar
EF_{NG} : Emission factor from LCA analysis of Natural gas powerplant
α: Land area per system capacity = 1.75439 ha/MWdc
12/44: conversion from CO2eq to C
A_{solar}: Area of solar panel
E_{SAM}: Annual electricity generation in 100MWdc utility-scale solar

Carbon Payback Time (CPBT)

$$CPBT = \min\left\{t \middle| \frac{\sum_t C_{LCA}}{\sum_t C_{saving}(t)} < 1\right\}$$

- Calculates the time required for a solar system to offset its life cycle carbon emissions
- Results are based on a representative 100 MWdc utility-scale solar installation (without deforestation)

Energy system carbon modeling results

<table>
<tr><th></th><th>Natural gas displacement (upper bound, static)</th><th>ISO-NE grid displacement (lower bound, time-variant)</th></tr>
<tr><td>2005 installation</td><td>
</td><td>
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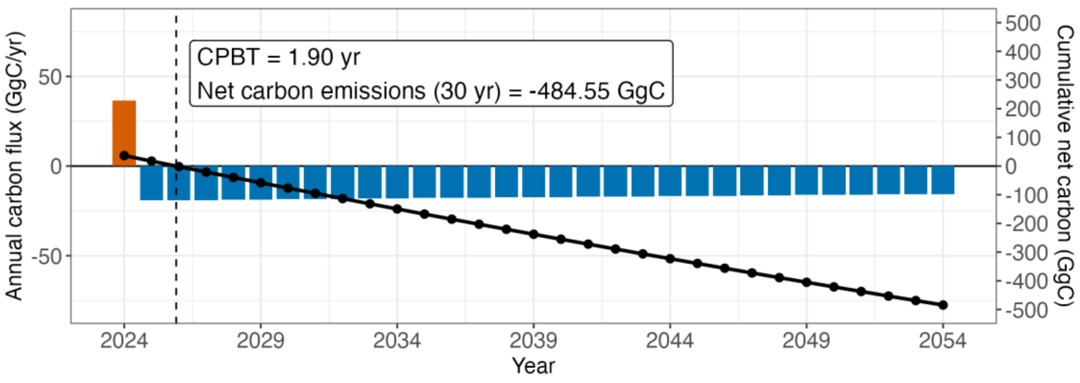
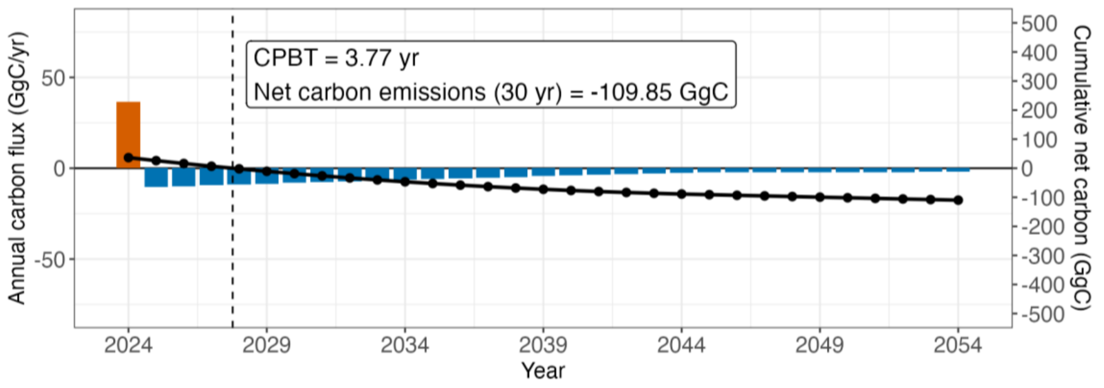
<tr><td>2024 installation</td><td>
</td><td>
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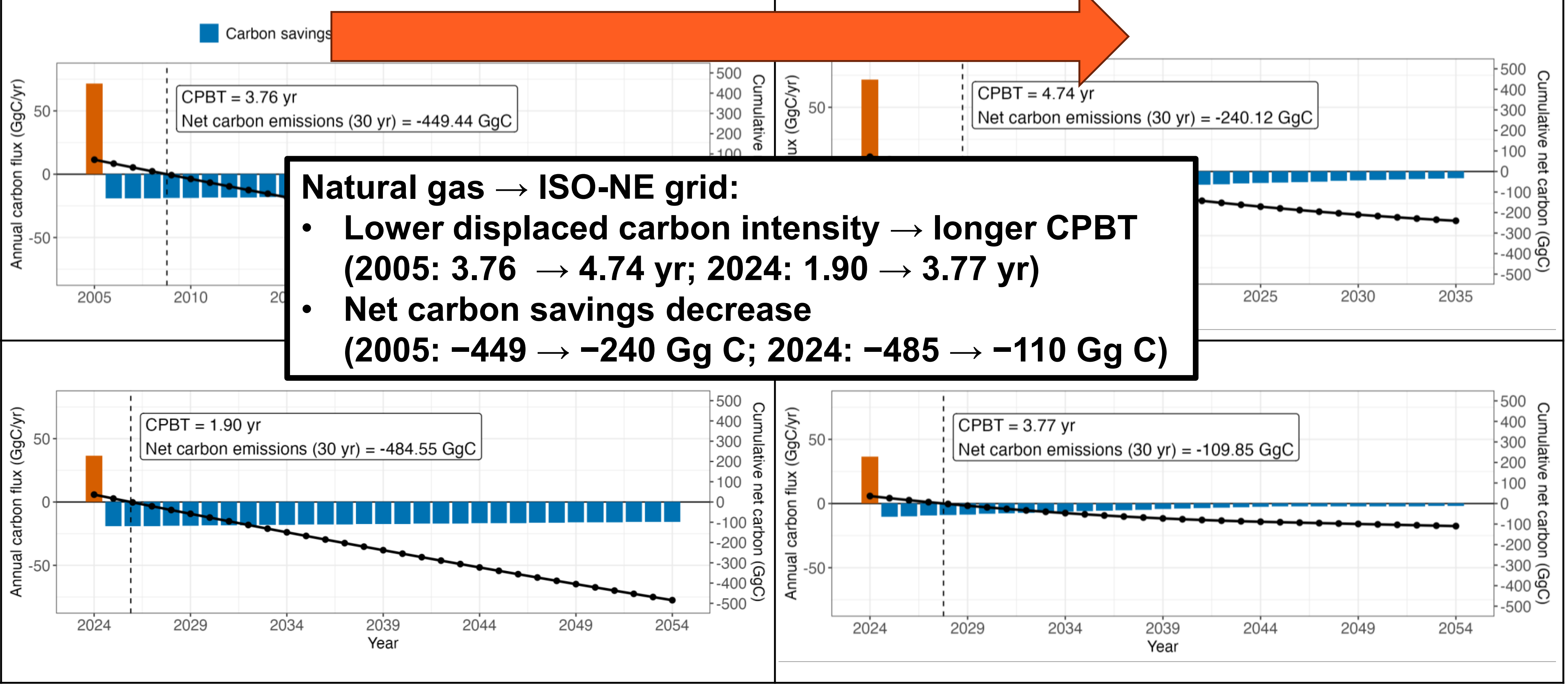
Energy system carbon modeling results

	Natural gas displacement (upper bound, static)	ISO-NE grid displacement (lower bound, time-variant)
2005 Installation		
2024 Installation		



Natural gas → ISO-NE grid:

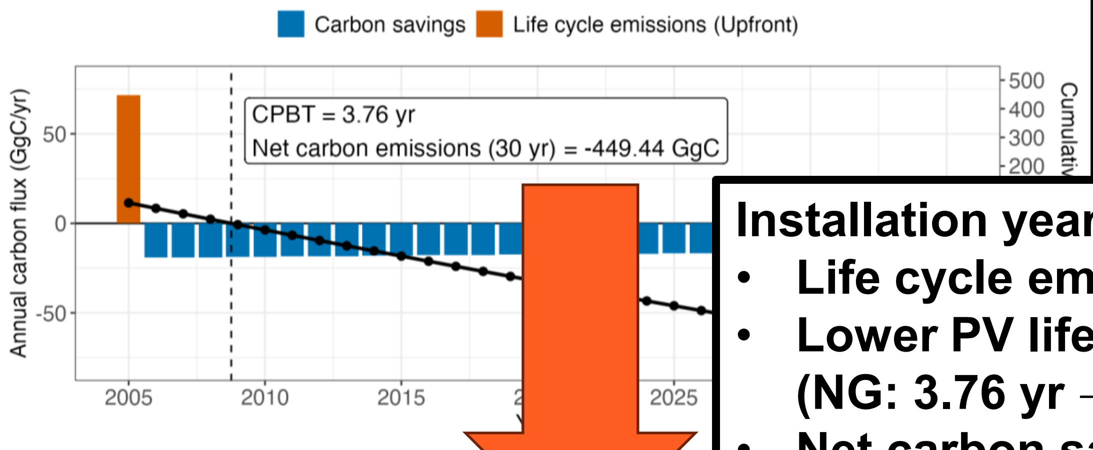
- **Lower displaced carbon intensity → longer CPBT (2005: 3.76 → 4.74 yr; 2024: 1.90 → 3.77 yr)**
- **Net carbon savings decrease (2005: −449 → −240 Gg C; 2024: −485 → −110 Gg C)**

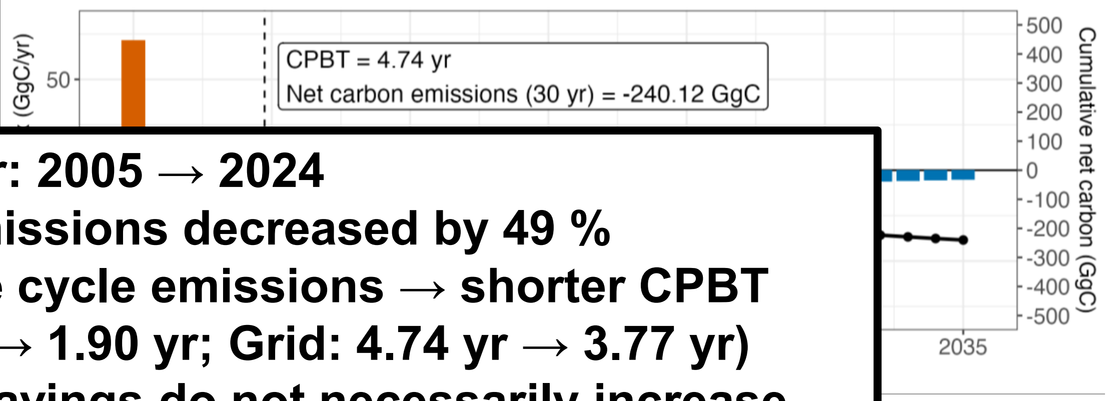
Energy system carbon modeling results

Natural gas displacement (upper bound, static)

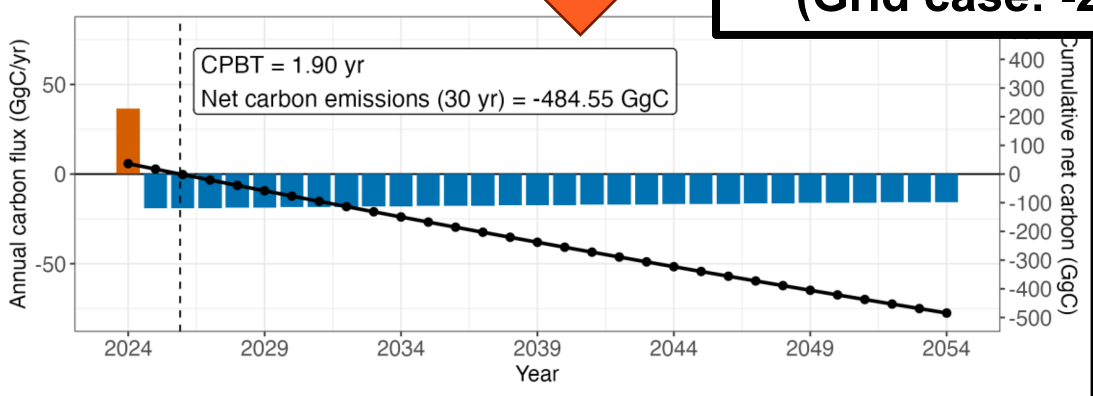
ISO-NE grid displacement (lower bound, time-variant)

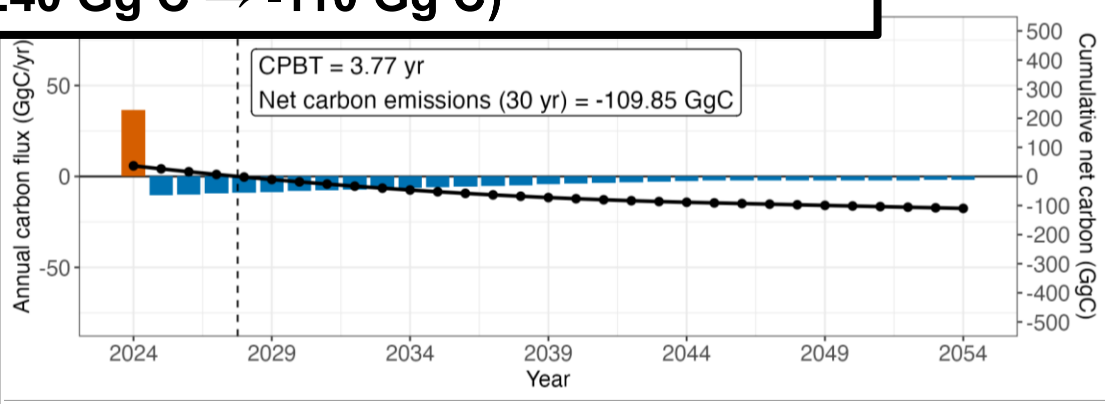
2005 installation





2024 installation





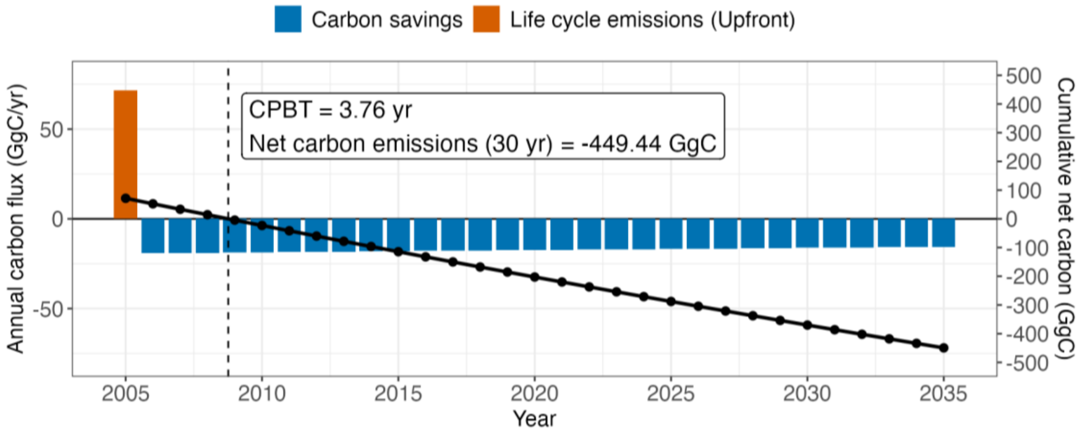
Installation year: 2005 → 2024

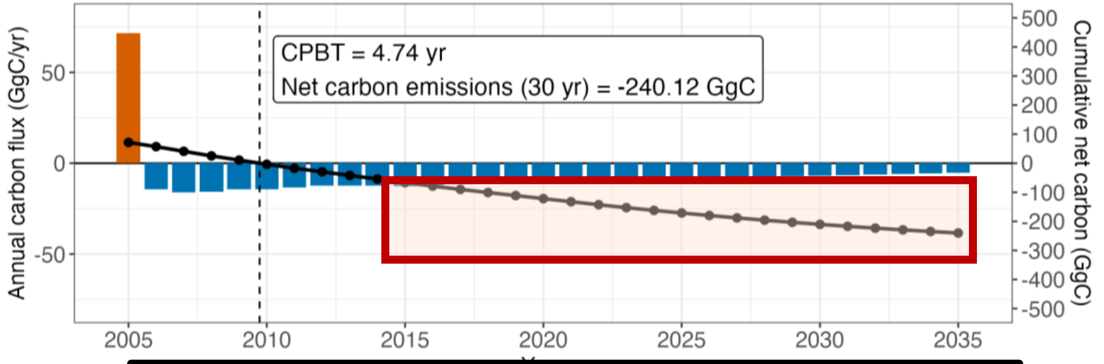
- **Life cycle emissions decreased by 49 %**
- **Lower PV life cycle emissions → shorter CPBT (NG: 3.76 yr → 1.90 yr; Grid: 4.74 yr → 3.77 yr)**
- **Net carbon savings do not necessarily increase (Grid case: -240 Gg C → -110 Gg C)**

Energy system carbon modeling results

Natural gas displacement (upper bound, static) | **ISO-NE grid displacement (lower bound, time-variant)**

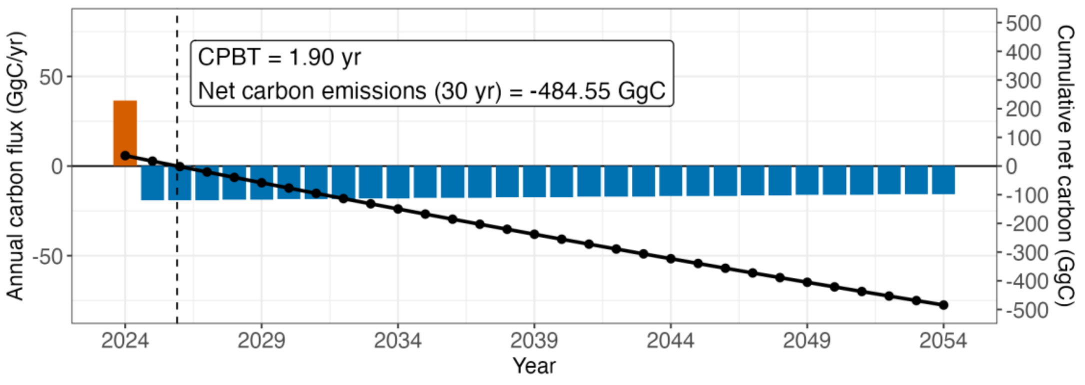
2005 Installation

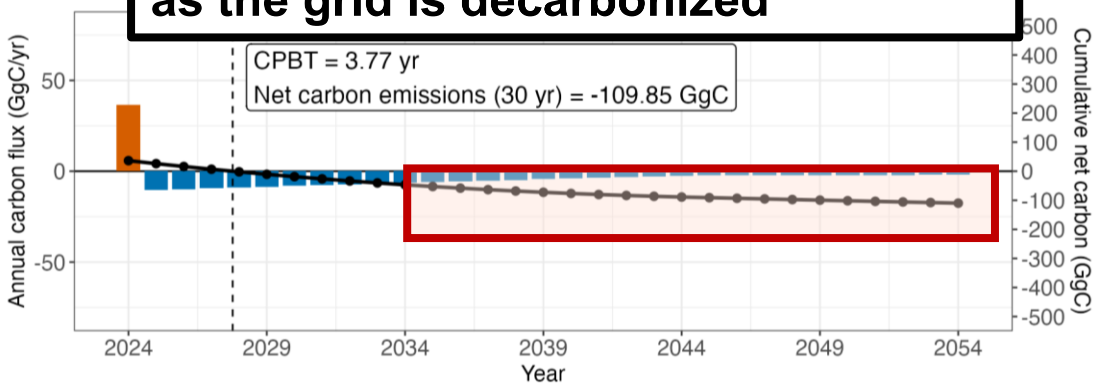




Annual net carbon benefits reduce as the grid is decarbonized

2024 Installation





Key Takeaways

$$C_{net}(t) = C_{Deforestation}(t) + \boxed{C_{LCA}(t) - C_{Saving}(t)}$$

- Under **natural gas scenario**, **lower PV life cycle emissions over time** lead to **shorter CPBT** and **larger lifetime carbon savings**
- Under **ISO-NE grid scenario**, **lower PV life cycle emissions** still **reduce carbon payback time**, but lifetime **net carbon savings decline** as displaced grid carbon intensity decreases
  CPBT estimates (1.9–4.7 years) are consistent with values reported in PV LCA literature (typically ~2–5 years)
- As the grid decarbonizes over time, the **reduction in carbon savings** becomes **larger than** the reduction in **PV life cycle emissions**
- As annual net carbon benefits reduce under grid decarbonization through time, **additional positive fluxes** lead to **increase in CPBT**

Chapter 3.

An Integrated Carbon Modeling with Uncertainty Propagation Linking Land Cover Changes, Energy Systems, and Biomass Dynamics

$$\boxed{C_{net}(t)} = \boxed{C_{Deforestation}(t)} + C_{LCA}(t) - C_{Saving}(t)$$

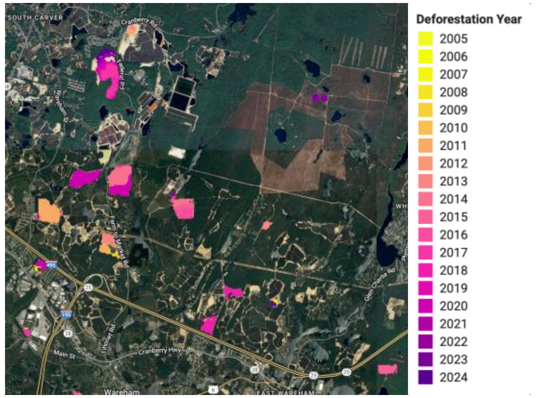
Deforestation carbon modeling

$$C_{net}(t) = \boxed{C_{Deforestation}(t)} + C_{LCA}(t) - C_{Saving}(t)$$

Carbon Emission from loss of Biomass

$$C_{AGB}(t) + C_{LoPS}(t)$$

Opportunity Cost: Carbon Loss from potential sequestration



Deforestation change map

Deforestation carbon

Deforestation area

Carbon emissions from deforestation $=$

$\times$

Annual AGB map

$\times$ Fraction of cleared biomass

$\times$ Decay rate

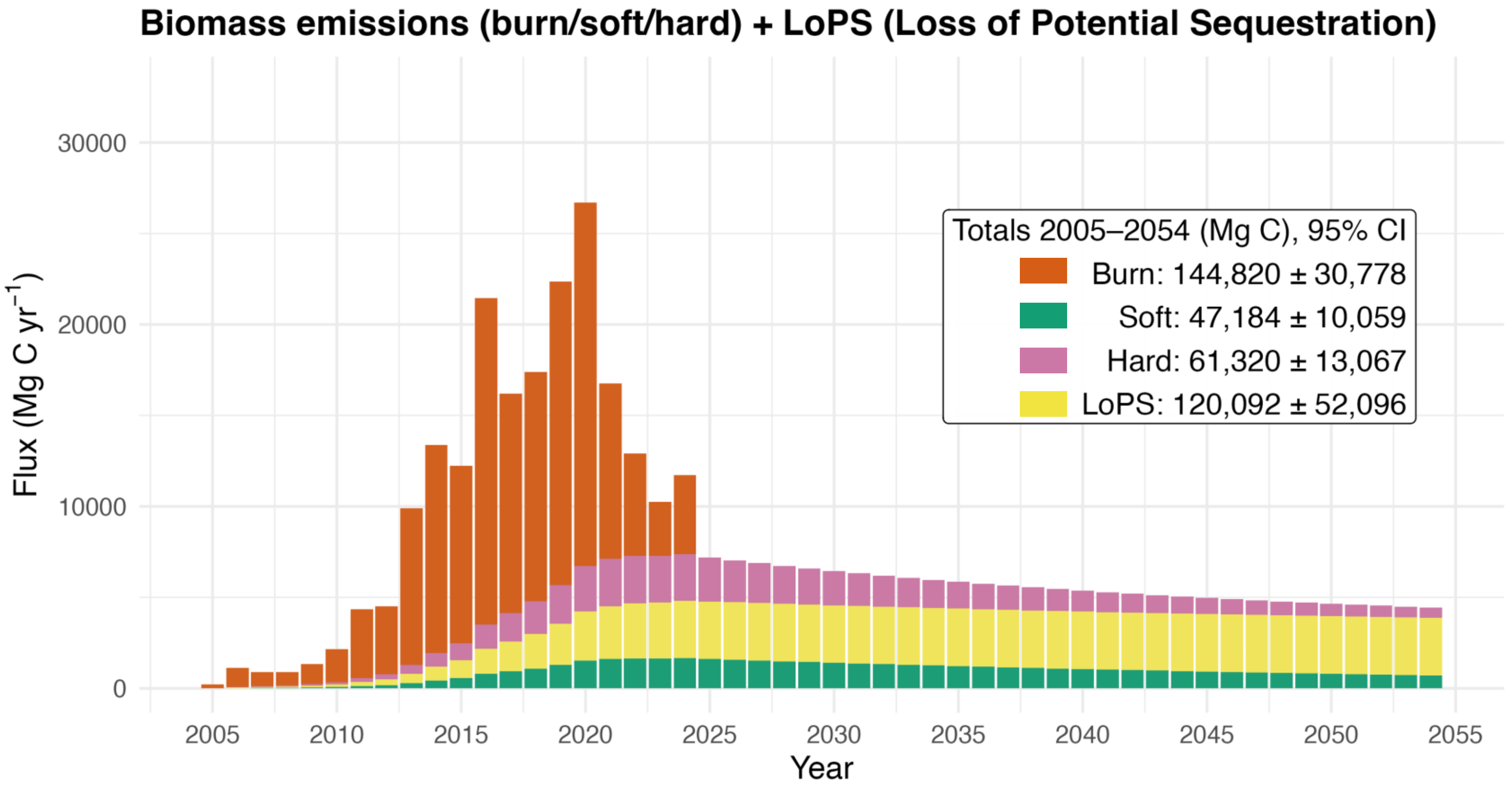
$1 - e^{d}$

Loss of potential carbon sequestration $=$

$\times$ Growth function

Carbon emissions from solar-driven deforestation

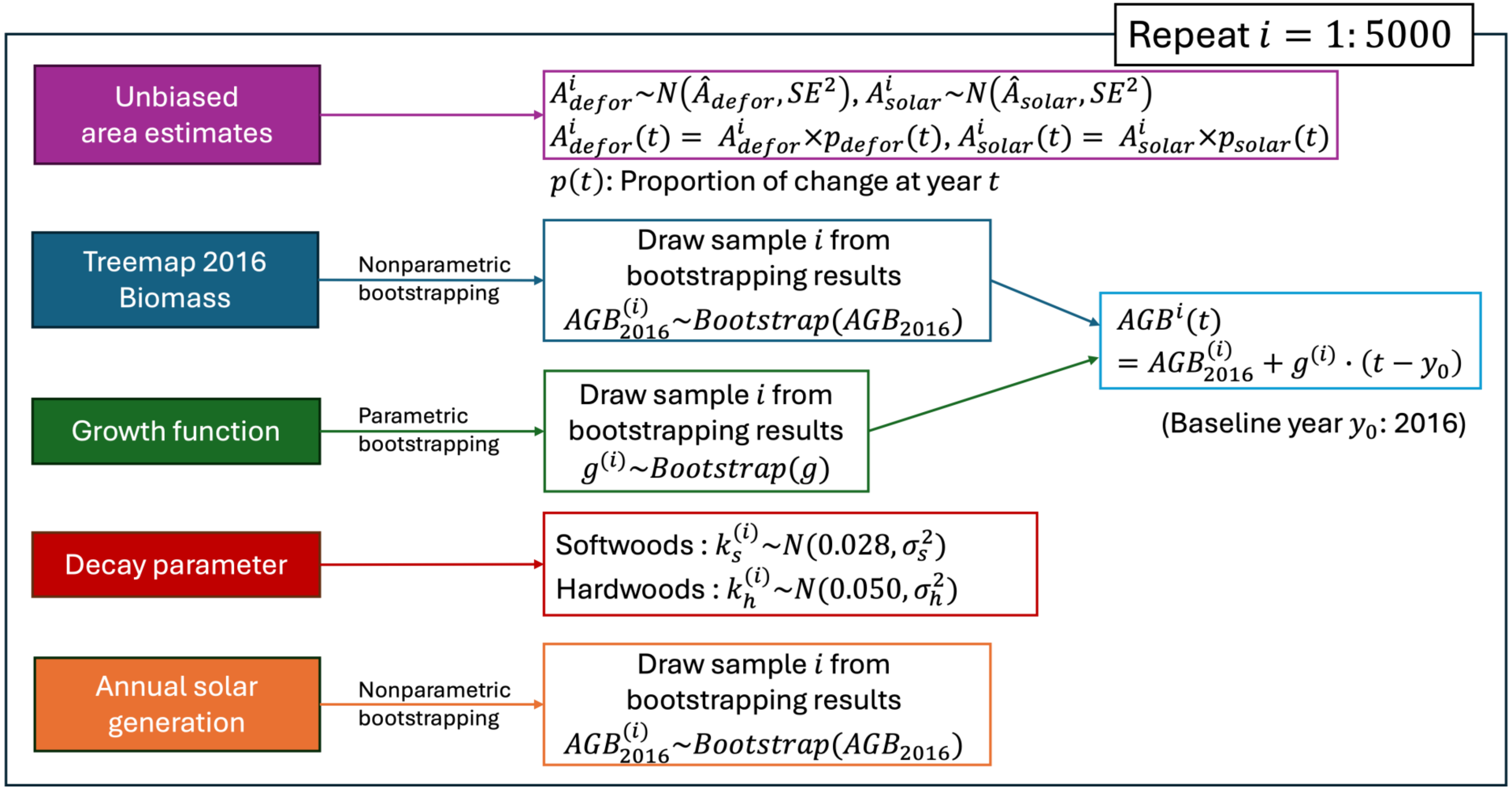
Components: combustion, decay, and foregone sequestration



- Peak emissions occur during rapid solar expansion (2013–2022)
- Loss of potential sequestration is a major long-term component
- Total deforestation emissions: **373 ± 91 Gg C**

Monte Carlo simulations of carbon modeling

$$C_{net}(t) = C_{Deforestation}(t) + C_{LCA}(t) - C_{Saving}(t)$$

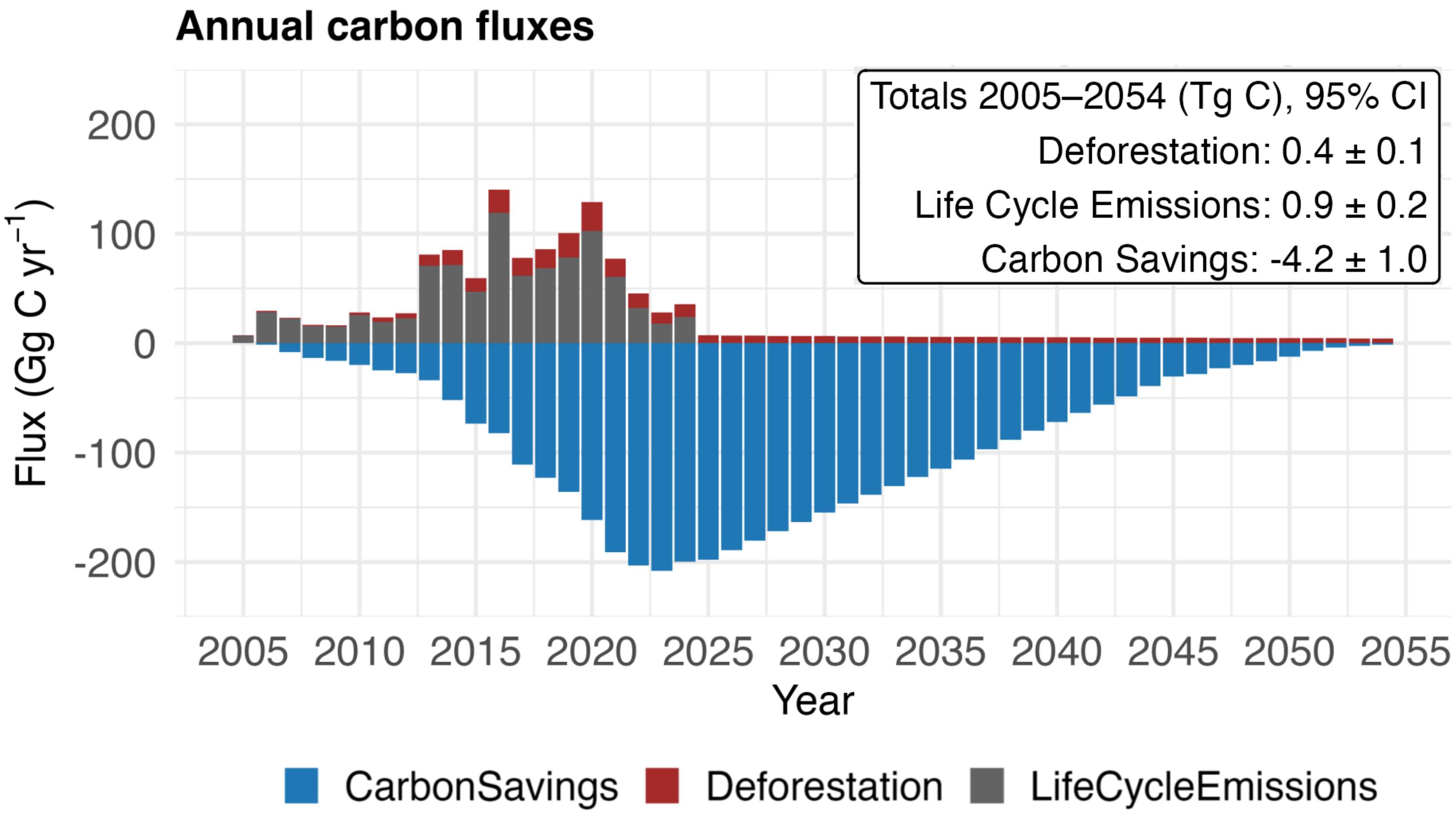


Simulation structure

- 5,000 Monte Carlo Simulations
- Combines spatial and available model uncertainty
- Produces uncertainty ranges for state-wide net carbon impact

Net carbon impact of statewide utility-scale solar development

Deforestation (Biomass) + LCA – Carbon savings (ISO-NE grid scenario)

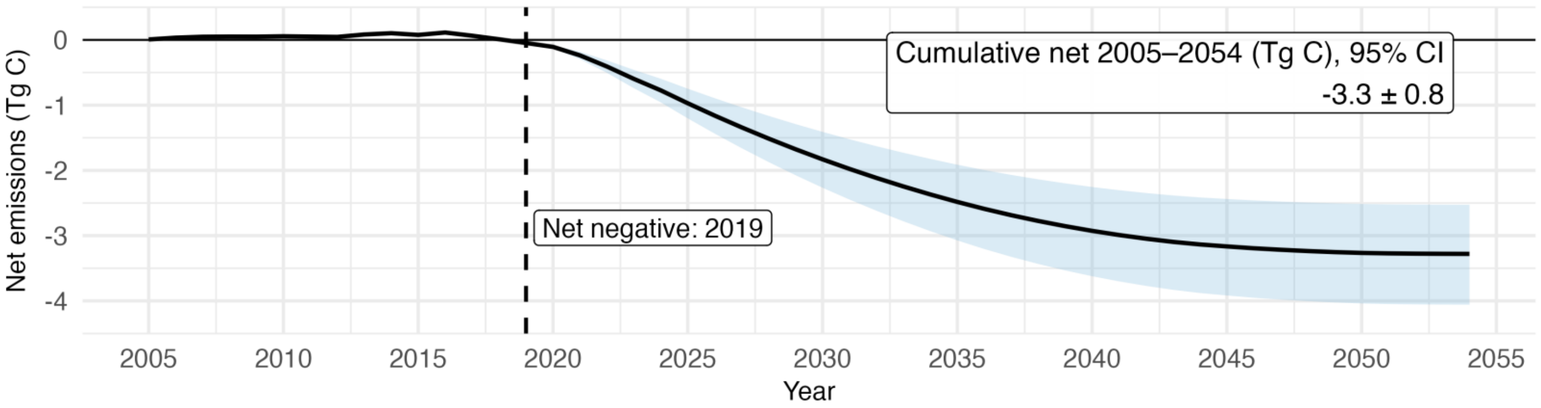


- Deforestation contributes 29% of total positive emissions
- Deforestation and life cycle emissions account for **22%** and **9%** of carbon savings
- Carbon savings (4.2 Tg C) are 3 times larger than all positive emissions (1.3 Tg C)

Net carbon impact of statewide utility-scale solar development

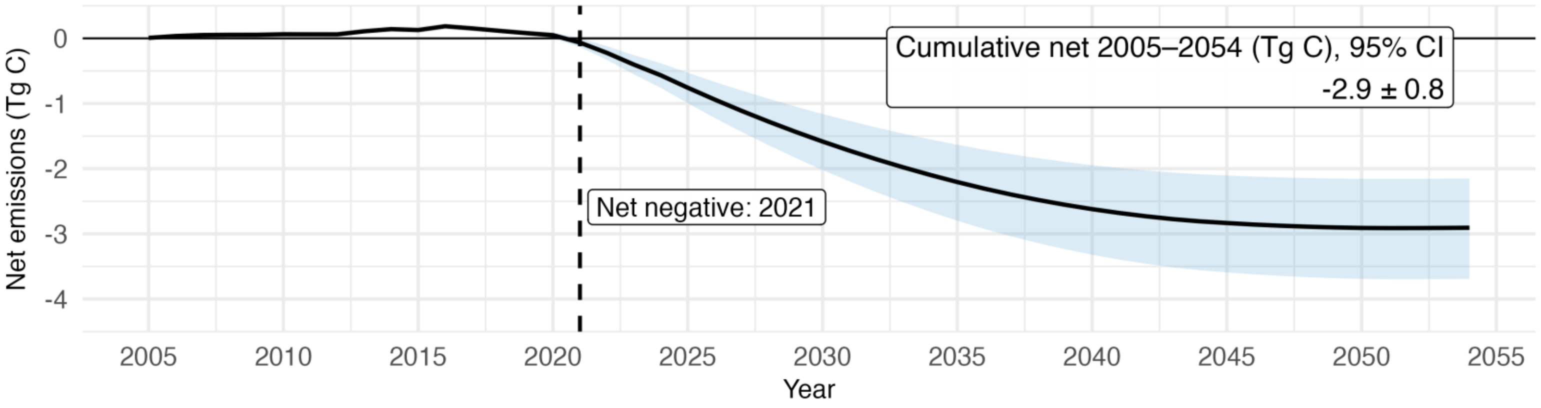
Cumulative net carbon emissions (w/o deforestation flux)

Mean with 95% CI (2005–2054)



Cumulative net carbon emissions (with deforestation flux)

Mean with 95% CI (2005–2054)

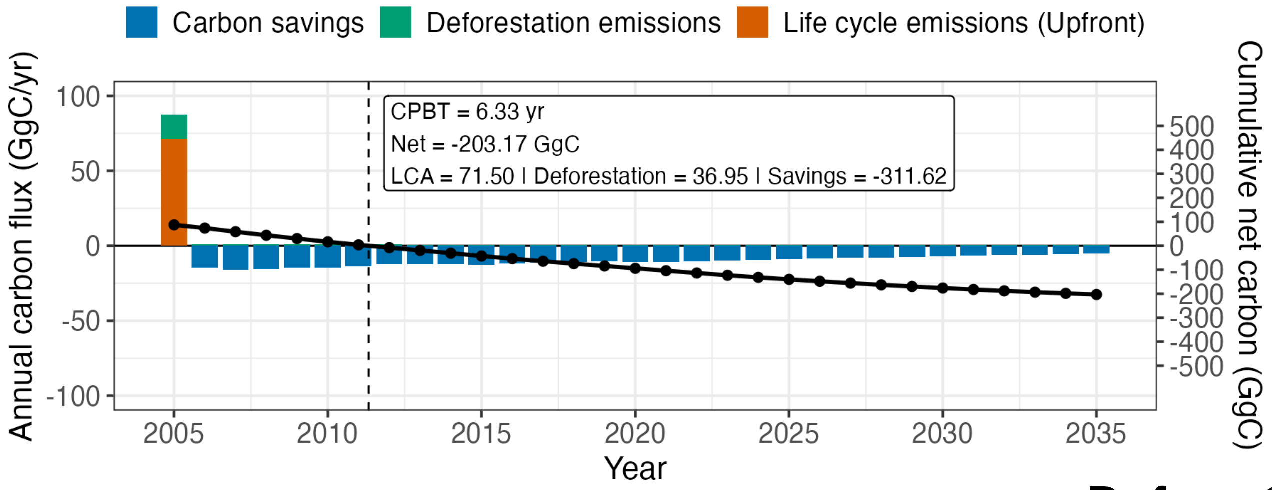


- Deforestation delayed statewide net carbon benefits by **2 years**
- Deforestation reduced carbon benefits of utility-scale solar systems by **12% (0.4 Tg C)**
- Long-term carbon benefits remain despite forest loss

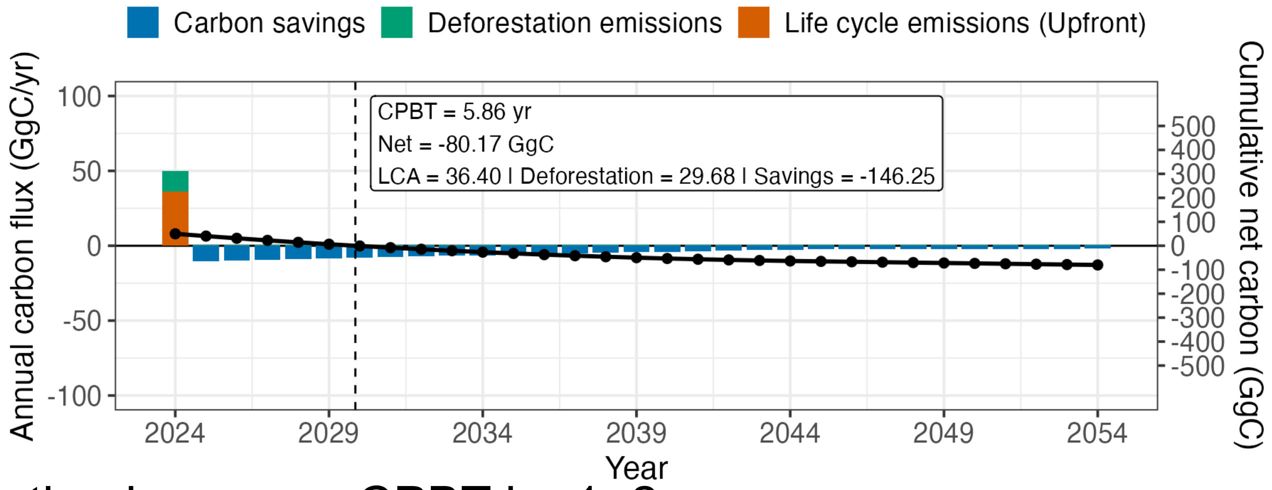
Effect of deforestation on solar carbon payback

Compare 2005 and 2024, ISO-NE grid displacement scenario, 100 MWdc with deforestation

Carbon Payback Time - Installed in 2005



Carbon Payback Time - Installed in 2024



Carbon Payback Time (CPBT)

$$CPBT = \min\left\{t \middle| \frac{\sum_t C_{LCA} + C_{Deforestation}}{\sum_t C_{saving}(t)} < 1\right\}$$

- Deforestation increases CPBT by 1~2 year
- Life cycle emissions grows from 22.9% to 24.9% of carbon savings
- Deforestation burden grows from **11.9% to 20.3%** of carbon savings over time
- The penalty of deforestation becomes larger as the grid get cleaner

Key takeaways

- Utility-scale solar in Massachusetts results in net carbon saving **(−2.91 ± 0.77 Tg C)** over 2005–2054
- Statewide carbon payback was reached around **2021**, with annual net carbon savings beginning in 2022
- Deforestation and life cycle emissions show **22%** and **9%** of total carbon savings over 2005–2054
- Despite smaller deforestation flux, its relative contribution to net carbon emissions increases over time **(from 11.9% to 20.3%)**
- Land use change and terrestrial carbon fluxes must be included for carbon modeling of utility-scale solar development

Summary and contributions

1. Developed a time-variant carbon modeling

- Incorporates temporal changes in PV life cycle emissions and grid carbon intensity
- Merging energy system and terrestrial carbon modeling

2. Net carbon impacts from utility-scale solar in Massachusetts

- Statewide carbon payback reached around **2021**
- Projected cumulative mitigations of **−2.91 ± 0.77 Tg C (2005–2054)**

3. Carbon savings decline over time due to grid decarbonization

- PV life-cycle emissions decrease over time
- However, declining grid carbon intensity reduces carbon savings even more

4. The relative importance of deforestation emissions increases over time

- Relative contribution of deforestation to net carbon emissions increases as grid system get decarbonized

Future directions

Expanding terrestrial carbon modeling beyond aboveground biomass

- Belowground biomass and soil carbon pools remain unquantified in this model and should be incorporated for more accurate carbon assessment

Improving estimates of operational carbon benefits of utility-scale solar

- Additional analysis needed to better reflect real-world carbon saving under evolving grid conditions

Scaling integrated carbon modeling beyond Massachusetts

- Integrated model can be extended to sub-continental scales, particularly across the eastern United States (Temperate Forest Climate)

Policy implications

Early solar installations provided net carbon benefits despite deforestation impacts

- Carbon savings from electricity generation relative to terrestrial carbon losses from deforestation decrease through time
- Solar development in MA (55 km^2) will save carbon equivalent to 1,064 km^2 of 30 years carbon uptake in MA

Land use carbon impacts become increasingly important as grids decarbonize

- Declining grid carbon intensity reduces avoided emissions while deforestation emissions remain constant

Sustainable siting will become increasingly important for future solar expansion

- Prioritizing disturbed or low-carbon lands for solar can reduce terrestrial carbon trade-offs

Carbon is only one dimension of utility-scale solar development

- Additional impacts, including biodiversity, forest fragmentation, microclimate effects, and ecosystem services, are not considered


Thank you for your attention